Seminar Presentation
on

Title: Supplier Selection using AHP and ANP: A Review and an Improved Model

Paper: Rodrigues, Širová & Mugurusi (2022), IFAC PapersOnLine 55-10, 2773–2778

by

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Seminar (Course Code: ME496)



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INTRODUCTION

- A small Czech company (about **400 workers**) makes parts for the car industry.
- Its machines need **copper rods** as a critical spare part. If a rod is missing, the **production line stops**.
- **Three suppliers** can sell the rod, and **three managers** (Maintenance, Production, Quality) must agree on one.
- Earlier, the choice was made by experience. The paper uses **AHP** and **ANP**, two scoring methods, to make it fair and explainable.

Problem: the two methods give **different winners**. AHP picks Supplier 2, ANP picks Supplier 1. This seminar checks **why** they differ and how the decision can be made **more reliable**.

Supplier	Location	What the paper says
Supplier 1	Germany	Best quality, highest price, farthest away
Supplier 2	Czech Republic (Central Bohemia)	Low price, fastest delivery
Supplier 3	Czech Republic (North Bohemia)	Low price, weakest quality

Source: Rodrigues et al. (2022), Sections 3–4. Price €230–350 per rod, delivery 3–7 days, bad parts 6.67–15% (2018).

LITERATURE REVIEW

Authors (Year)	Title	Research Point
Saaty (1980)	The Analytic Hierarchy Process, McGraw-Hill, New York.	• Breaks a decision into a tree of criteria • Compares two items at a time on a 1–9 scale • Weights come from the eigenvector • Consistency ratio checks the answers
Saaty (2005)	Theory and Applications of the Analytic Network Process, RWS Publications.	• Network instead of a tree (ANP) • Criteria can affect each other • Uses a supermatrix for feedback
Alegoz & Yapicioglu (2019)	Supplier selection and order allocation decisions under quantity discount and fast service options, Sustainable Production and Consumption, 18, 179–189.	• Picks suppliers and splits the order • Uses both cost and service criteria • Includes fast-delivery options
Rodrigues, Šírová & Mugurusi (2022)	A supplier selection decision model using multi-criteria decision analysis in a small manufacturing company, IFAC PapersOnLine, 55(10), 2773–2778.	• Base paper of this seminar • AHP vs ANP for a copper rod supplier • 3 managers, 6 attributes, 3 suppliers • The two methods give different winners

METHODOLOGY OF THE PAPER

1. Six attributes in **two groups**:
   a. Logistic-Economic: delivery time, price, availability
   b. Quality-Sustainability: quality, aftersales, sustainability
2. Managers compare **two items at a time** on Saaty's **1–9 scale**.
3. Three answers combined by **geometric mean**.
4. **SuperDecisions** software computes the scores.
5. Solved twice: **AHP** (tree) and **ANP** (network).

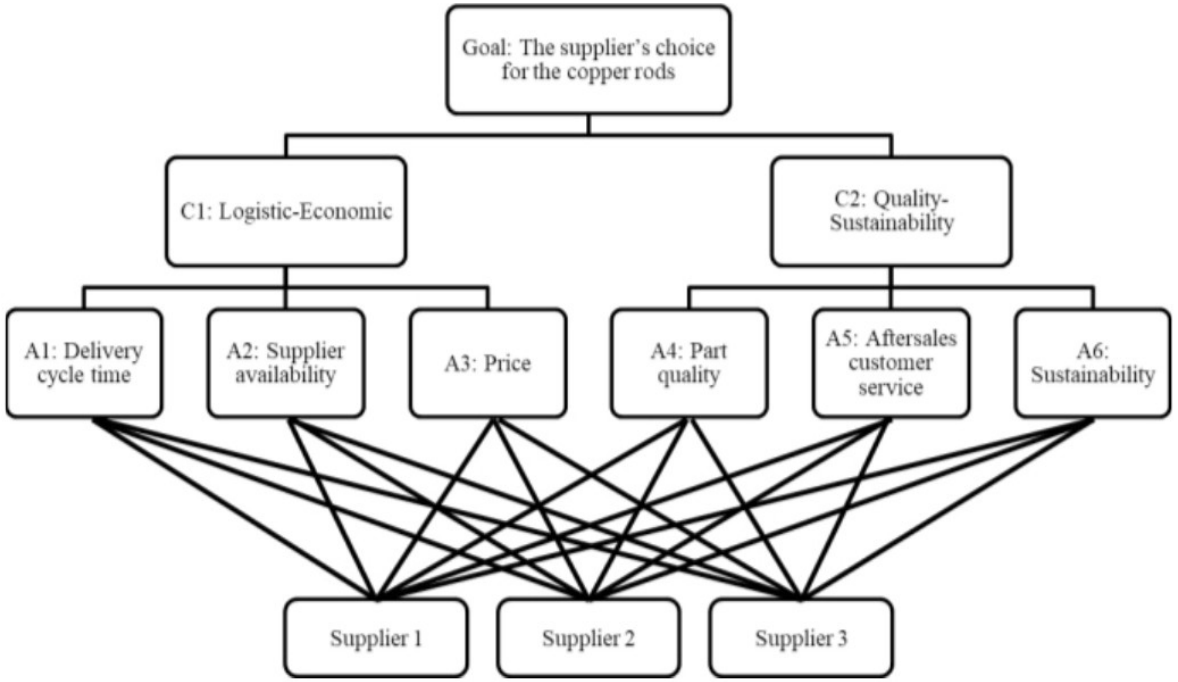


Fig. 1: Hierarchical tree of the decision

Source: Rodrigues et al. (2022), Fig. 1, p. 2775

Paper's result	S1	S2	S3	Winner
AHP	0.371	0.419	0.209	Supplier 2
ANP	0.546	0.319	0.135	Supplier 1

Source: Rodrigues et al. (2022), Tables 10–11

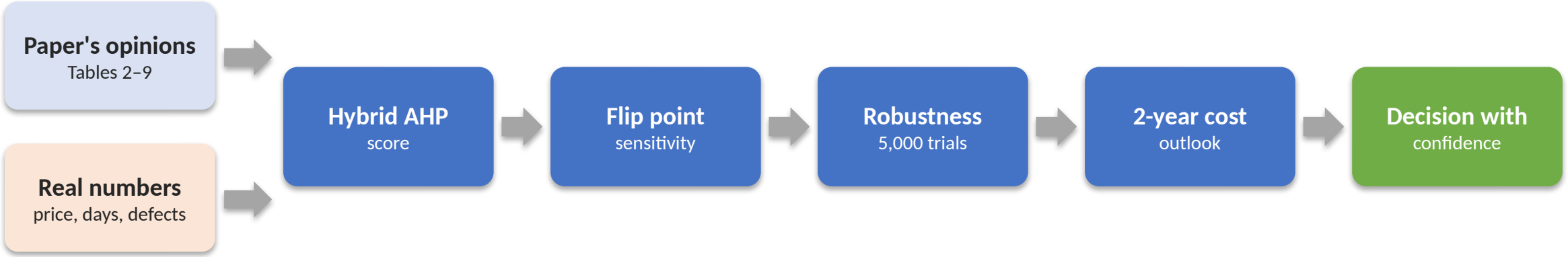
RESEARCH GAPS

Gap in the paper	What I found	My Approach
ANP gives a different winner	ANP result = AHP with groups weighted 1 : 5. One opinion changes the answer.	Show the exact flip point to the managers
No sensitivity check	Supplier 1 wins below 0.44 Logistic weight, Supplier 2 above it.	What-if slider and one-attribute sensitivity table
Real numbers turned into opinions	Price only 1.5 times higher, but scored about 7 times worse.	Score price, delivery days and defects from real numbers
Inconsistent answers	4 of 8 tables are above Saaty's 5% limit for three items.	Live consistency check while entering answers
One fixed ranking	Opinions are never exact, so one ranking hides the risk.	5,000 random trials: % of times each supplier wins
No long-term view	Ranking does not show cost of bad parts or line stops.	2-year cost outlook for each buying plan

Source: own re-analysis of Rodrigues et al. (2022), Tables 2–12 (calculations in Excel workbook and demo)

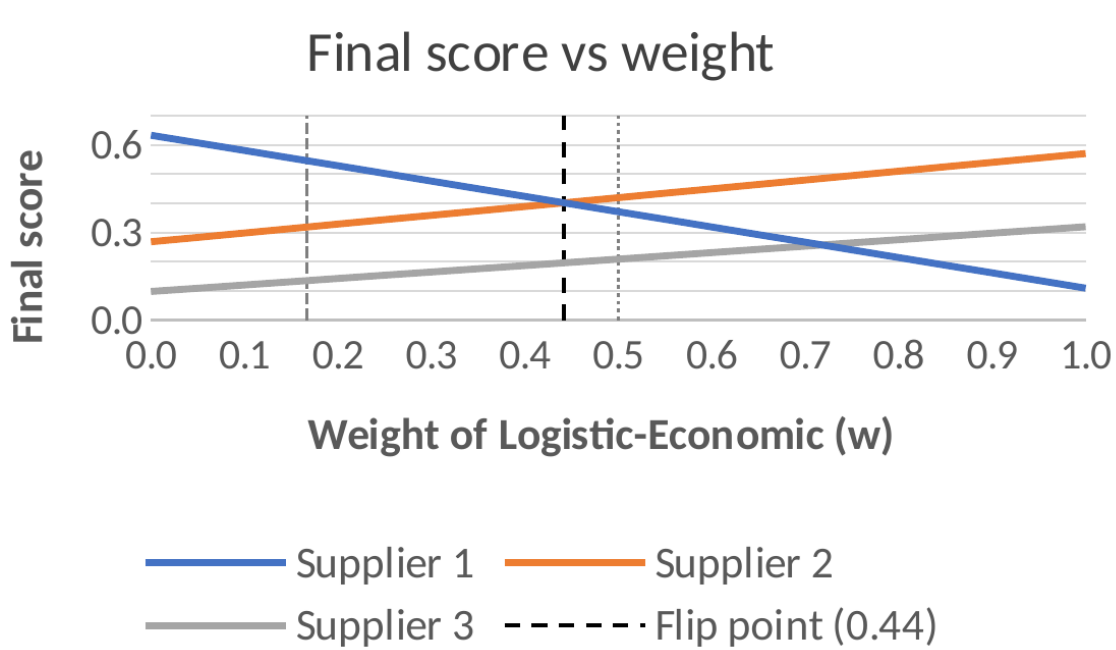
PROPOSED IMPROVEMENT

1. **Rebuild and check** the paper's model
   a. Recompute AHP and ANP from the paper's own tables and check consistency.
2. Use **real numbers** where they exist
   a. Price, delivery days and defect rate scored as 1 / value, not as opinions.
3. Measure **how sure** the answer is
   a. Find the flip point, then run 5,000 random trials (Monte Carlo).
4. Show the **long-term effect** of the choice
   a. Simulate 24 months of buying, bad rods and line stops, including a split order.



Built as an offline web demo (HTML, JavaScript, Chart.js). Runs with one command: node server.js

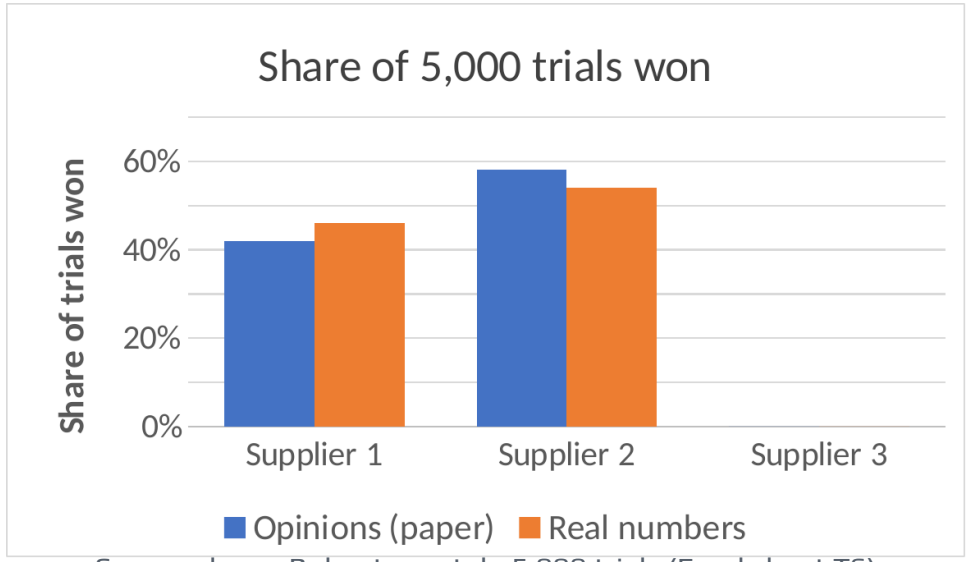
RESULTS AND DISCUSSION



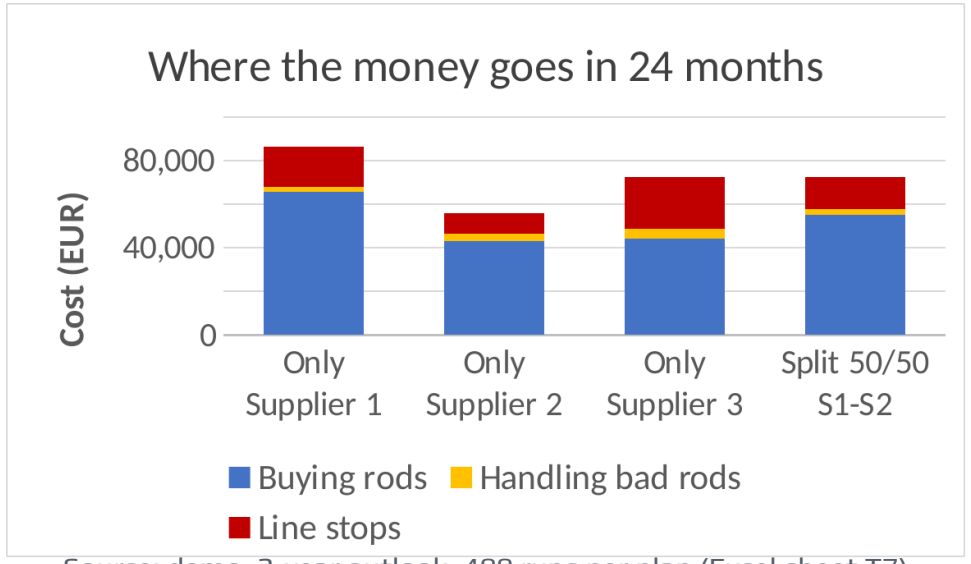
Dotted lines: paper ANP (0.17), AHP (0.50). Source: own calculation, paper Tables 2–9 (Excel sheet T3)

- Rebuilt AHP and ANP match the paper to **4 decimals**.
- Supplier 3 **never wins**. S2 vs S1 is close: **58% / 42%** of trials.
- 2-year cost: S2 about **€56k**, S1 about **€86k**. S1 pays off only if one bad rod costs over **€5,250**.

Demo ↗



Source: demo, Robustness tab, 5,000 trials (Excel sheet T6)



Source: demo, 2-year outlook, 400 runs per plan (Excel sheet T7)

CONCLUSIONS AND FUTURE WORK

Conclusions

1. **Same model, one opinion:** the AHP and ANP answers differ only because quality was made 5 times more important.
2. **Close call:** S1 and S2 are nearly tied. A single ranking hides this.
3. **Better advice:** drop S3, and choose S2 for cost and speed, or S1 only if bad parts are very costly. A split order is a safe middle path.

Future Work

- ☐ Use **fuzzy AHP** to handle unsure answers.
- ☐ Give managers **different weights** by experience.
- ☐ Use the company's **real yearly data** in place of estimates.
- ☐ Find the **best split** of orders with optimisation.
- ☐ Connect the demo to **live ERP data** so scores update by themselves.

"A good decision model should not only say who wins, but also how sure we are and what it will cost later."